

Task Name

Dynamic Contextual Pricing Engine with Multi-Armed Bandits (Thompson Sampling)

Objective

The objective of this project is to build a dynamic pricing simulation that automatically adjusts product prices based on customer behavior and market conditions. The system uses the **Thompson Sampling** algorithm, a Multi-Armed Bandit technique, to identify the most profitable price while balancing exploration and exploitation.

Problem Statement

Businesses often use fixed prices even though customer demand changes continuously. Factors such as customer interest, competitor prices, and buying behavior influence sales. Selecting the best price manually is difficult because customer preferences keep changing.

This project develops a dynamic pricing engine that learns from customer responses and updates prices automatically to maximize revenue.

Concept Used

1. Dynamic Pricing

Dynamic pricing means changing product prices according to demand and market conditions.

Example:

- Morning Price = ₹450
- Afternoon Price = ₹500
- Evening Price = ₹470

The system continuously decides which price performs better.

2. Multi-Armed Bandit

The Multi-Armed Bandit problem represents several possible choices.

Each price acts as one "arm."

Example: The algorithm chooses one price at a time and learns from customer purchases.

Arm	Price
Arm 1	₹ 400
Arm 2	₹ 450
Arm 3	₹ 500
Arm 4	₹ 550
Arm 5	₹ 600

3. Thompson Sampling

Thompson Sampling is a probabilistic algorithm.

For every price, the algorithm stores:

- Number of successful purchases
- Number of unsuccessful visits

Each price has a Beta probability distribution:

Beta(Success + 1, Failure + 1)

The algorithm randomly samples from each Beta distribution and selects the price with the highest sampled value.

This method naturally balances:

- Exploration (trying new prices)
- Exploitation (using successful prices)

Algorithm

Step 1

Initialize different price options.

Example:

400
450
500
550
600

Step 2

For every price initialize

Success = 1

Failure = 1

Step 3

Observe customer interaction.

Customer purchases product:

Success += 1

Otherwise

Failure += 1

Step 4

Calculate Beta Distribution

Beta(Success + 1, Failure + 1)

Step 5

Sample one value from every Beta distribution.

Step 6

Choose the price with the highest sampled value.

Step 7

If competitors reduce prices, simulate a competitive shock.

Example:

Competitor price drops from ₹500 to ₹450.

The algorithm immediately explores lower prices before settling on the best-performing option.

Price	Success	Failure	Conversion Rate
₹400	30	20	60%
₹450	42	18	70%
₹500	55	15	79%
₹550	21	32	40%
₹600	10	38	21%

Simulation ExampleInitially, the algorithm explores all prices.

After learning customer behavior, it selects ₹500 more frequently because it provides the highest revenue and conversion probability.

Competitive Shock Example

Initially

Competitor Price

₹520

Our Price

₹500

Later competitor reduces price

₹450

The algorithm temporarily explores

₹450

₹470

₹480

After collecting new customer responses, it again converges to the most profitable price.

Expected Output

- Dynamic price recommendations
 - Continuous learning from customer purchases
 - Better revenue optimization
 - Fast adaptation to competitor price changes
 - Balanced exploration and exploitation
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Applications

- E-commerce websites
- Airline ticket pricing

- Hotel booking platforms
 - Online food delivery
 - Ride-sharing services
 - Retail stores
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Advantages

- Maximizes revenue
 - Learns automatically from customer behavior
 - Reduces manual pricing decisions
 - Responds quickly to market changes
 - Improves customer conversion rate
 - Easy to scale for large businesses
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Conclusion

The Dynamic Contextual Pricing Engine demonstrates how Multi-Armed Bandits and Thompson Sampling can intelligently optimize product prices. By continuously learning from customer conversion data and reacting to competitive market changes, the system improves pricing decisions while maximizing revenue. This approach provides an efficient and adaptive solution for modern businesses operating in dynamic markets.